

PERSONAL INFORMATION

Ahcene Aliouet - Associate Professor (Maître de conférences)
INSA Hauts-de-France & Université Polytechnique Hauts-de-France,
Valenciennes, France
 [Google Scholar Profile](#)
 [Github Profile: @ahcen23](#)
 [LinkedIn Profile: www.linkedin.com/in/ahcenaliouat](#)
 [ORCID: https://orcid.org/0000-0001-7590-9532](https://orcid.org/0000-0001-7590-9532)



KEYWORDS :

Goal-oriented and Semantic Communications ; Image compression for computer vision tasks ; Image Communication ; Image Processing ; Wireless Communication ; Deep Learning ; Machine Learning ; Signal Processing ; Image Compression ; Video Coding ; Object Detection ; Region-of-Interest Detection, Information Theory, Rate-Distortion, Rate-Distortion-Perception

EMPLOYMENT HISTORY

Associate Professor - Maitre de Conférences INSA HdF/IEMN Lab *September 2025-present*

Teaching : Signal Processing, Image and Video coding and compression, Signal Processing and Filters, Image Processing and Advanced Image Processing courses. for INSA HdF Audiovisuel Engineering students, and also License 3rd year, and Master's students of the UPHF university. On the research side, working on video coding for machine standard (VCM). Learning for Coding and Coding for Learning, also developing new compression techniques for learning on compressed data without any decoding.

Postdoctoral Researcher MEE Department, IMT Atlantique

July 2023-present

Learning on coded/compressed data. We are applying and enhancing Deep Learning methods for learning on compressed and coded data(images/video) without (or with partial) decoding (source + channel coded data) by developing, among others, learnable decoders (neural belief propagation) in Goal-Oriented communications applications to compressed-domain representation and image retrieval. Working with Elsa Dupraz as Principal Investigator, Pr. François Xavier Socheleau on the work related to sub-marine acoustic signal compression and classification, and Pr. Aline Roumy for the work on Image retrieval on compressed data, without decompression.

Part-time Lecturer IMT Atlantique

Mars 2025- Present

Supervision of the projects related to the teaching unit "Data Compression : From source coding to virtual reality"

- Video compression using common standards (JPEG, MPEG, ...)
- Intra. and Inter-coding
- Predictive coding
- Rate-Distortion evaluation

Part-time Lecturer University of Brest (UBO), Electronics Department, Parcours Signal et Telecom *September 2024-February 2025*

Teaching of the Integration Project Module (40 hours), focusing on machine learning for visual tasks.

Module Outline :

- 12 hours : Introduction to Machine and Deep Learning
 - a) Fundamentals of machine learning
 - b) Convolutional neural networks
 - c) Various loss and activation functions
 - d) Calculation of partial derivatives for backpropagation
 - e) Overview of different neural network architectures (RNN, LSTM, GRU, autoencoders, etc.)
- 25 hours : Project Proposals and Supervision
 - a) Proposal of seven projects, one project per pair of students
 - b) Projects encompass various visual tasks, including classification of medical images (e.g., chest X-rays, skin cancer detection) ; Face recognition ; Image denoising using autoencoders ; Analysis of the impact of image compression on classification.
- 3 hours : Evaluation of Reports and Oral Presentations
 - a) Assessment of written project reports
 - b) Final oral defenses by students.

Part-time Lecturer IMT Atlantique*September 2024- December 2024*

Labs on Machine Learning for 2nd year engineering students : practical sessions in "Introduction to Machine Learning", which includes four labs :

- Convolutional Neural Networks (CNN) - 3 hours
- Polynomial Regression - 4 hours and 15 minutes
- Principal Component Analysis (PCA) - 1 hour and 30 minutes
- Support Vector Machines (SVM) - 3 hours

Part-time Lecturer Badji Mokhtar - Annaba University*October 2019-June 2021*

Labs in Algorithmics and Programming 36 hours for first-year Science and Technology students, covering :

- Algorithmics (12h)
- Mathematical Methods (12h)
- Matlab Programming (12h)

Visiting Ph.D. Student (PHC TASSILI Cotutelle scholarship) CRAN Lab., Lorraine university*October 2022-November 2022*

Synthesizing and writing a first draft of a journal paper on background subtraction for moving object detection (have been published in 2023). Revision of a second journal paper on multi-level ROI detection using the proposed SSAD method (published in Q1 journal at the end of 2022).

Visiting Ph.D. Student (PHC TASSILI Cotutelle scholarship) CRAN Lab., Lorraine university*November 2021-January 2022*

Working on an implementation of a proposed region-of-interest (ROI) detection method in SensVid, a dedicated platform for video coding and transmission in wireless sensor networks. This method was integrated as an additional option to improve encoding performance by preserving crucial information and reducing energy and bitrate consumption. The development was carried out using OpenCV and C++.

EDUCATION

Doctorate in Multimedia and Digital Comm. LASA, Badji Mokhtar - Annaba University *Mai 2023*

- **Thesis title** : Study and implementation of an object-based video encoder for embedded wireless surveillance systems.
- **Context** : The thesis addresses the challenge of preserving high-quality video/image transmissions in battery-powered wireless multimedia sensor networks for surveillance applications by devising novel, energy-efficient encoding methods. A particular emphasis is placed on region-of-interest (ROI) detection for coding, and evaluated based on Quality of Experience (QoE) maintenance for both human visual systems and machine-based systems (AI models). The work systematically evaluates its multiple proposed approaches using extensive performance indicators -including classification metrics (F-score, Recall, Specificity, Precision, PWC, BAC) and image quality metrics (PSNR, SSIM, VIF, BRISQUE)- and demonstrates improved energy consumption and transmission quality over existing solutions, thus offering a viable strategy for resource-constrained environments.
- **Outputs** : 4 journal papers (with IF > 3) and 2 international conference papers. Awarded best thesis at the University of Annaba Badji Mokhtar in 2023.

Master Degree in Telecommunications, Networks and, Multimedia University of Sciences et de la Technology Houari Boumediene (USTHB) *September 2016 - July 2018*

Title : Non-Orthogonal Multiple Access (NOMA) combined with Random Linear Network Coding (RLNC) in Cooperative Communication

Supervisor : Pr. Fatiha Merazka

Licence Degree in Telecommunications and Computer Networks Engineering University of Sciences et de la Technology Houari Boumediene (USTHB) *September 2013 - June 2016*

Project Title : Simulation of a Nakagami-m Fading Channel

Supervisor : Pr. Fatiha Merazka

SKILLS

Operating systems : Windows and Linux.

Programming languages : MATLAB, Python, C, C++.

Office softwares : Microsoft Office, LaTeX, Git and Github, Overleaf.

Scientific softwares MATLAB, Deep Learning Toolbox, Image Processing Toolbox, PyTorch, TensorFlow, OpenCV.

DL engineering approaches : RNN/GRU/LSTM/CNN/Regression/SVM/PCA.

System Level : OpenVino by Microsoft Toolkit, Cisco Packet Tracer.

Circuit Level : VHDL FPGA (basics).

Languages : French, English, Arabic.

AWARDS

- **Best Thesis Award :** 15 July 2023
Badji Mokhtar - Annaba University, out of 11 engineering departments
*Thesis Title : **Study and implementation of an object-based video encoder for embedded wireless surveillance systems.***

the thesis is available here :

<https://biblio.univ-annaba.dz/wp-content/uploads/2024/02/These-Aliouat-Ahcen.pdf>

Supervised by :

- Pr. Nasreddine Kouadria (Badji Mokhtar - Annaba University)
- Pr. Saliha Harize (Badji Mokhtar - Annaba University)
- Dr. HDR. Moufida Maimour (University of Lorraine, CRAN Laboratory)
- Pr. Nouredine Doghmane (Badji Mokhtar - Annaba University)

PROFESSIONAL ORGANIZATIONS

The Institute of Electrical and Electronics Engineers (IEEE), Member

- IEEE Membership 2022-present
- "IEEE Young Professionals" society 2022-present

European Association For Signal Processing (EURASIP), Regular Member 2024-present

PUBLICATIONS

Under Review :

- [Conf1] | "Task-Oriented Source Coding Using LDPC Codes for Compressed-Domain Image Retrieval", *International Conference on Image Processing (ICIP) 2026*. Mars 2026. **Authors : Ahcene Aliouet, Yann Miguet, Elsa Dupraz, Aline Roumy.**

Main Researches

- [J1] | "Goal-Oriented Source Coding using LDPC Codes for Compressed Domain Image Classification", (Under Review) in *IEEE Transactions on Communications*
Authors : Ahcen aliouat, Elsa Dupraz
- [C1] | "Apprentissage sur des images compressées JPEG-LDPC : Classifier les syndromes", in *COmpression et REpresentation des Signaux Audiovisuels (CORESA 2024)*, Rennes, France.
Authors : Ahcen aliouat, Elsa Dupraz
- [C2] | "Learning on JPEG-LDPC Compressed Images : Classifying with Syndromes", in *EUSIPCO 2024 Conference*, Lyon, France.
Authors : Ahcen aliouat, Elsa Dupraz

- [J2] | "EVBS-CAT : Enhanced Video Background Subtraction with a Controlled Adaptive Threshold for Constrained Wireless Video-surveillance", Ahcen Aliouat & al., in *Journal of Real-Time Image Processing (Springer)*
Authors : Ahcen aliouat, Nasreddine Kouadria, Moufida Maimour, Saliha Harize
- [J3] | "x-DTT : A Package for Calculating Real and Integer Discrete Tchebichef Transform Kernels based on Orthogonal Polynomials", Ahcen Aliouat & al., in *SoftwareX (Elsevier)*
Authors : Ahcen aliouat, Nasreddine Kouadria,
- [J4] | "An Efficient Low Complexity Region-of-Interest Detection for Video Coding in Wireless Visual Surveillance", Ahcen Aliouat & al., in *IEEE Access*
Authors : Ahcen aliouat, Nasreddine Kouadria, Saliha Harize, Moufida Maimour
- [J5] | "Region-of-Interest based Video Coding Strategy for Rate/Energy-constrained Smart Surveillance Systems using WMSNs", in *Ad Hoc Networks (Elsevier)*
Authors : Ahcen aliouat, Nasreddine Kouadria, Saliha Harize, Moufida Maimour, Norededdine Doghmane
- [C3] | "Region-of-Interest based Video Coding Strategy for Low Bitrate Surveillance Systems", Ahcen Aliouat & al., in *IEEE 19th International Multi-Conference on Systems, Signals and Devices (SSD'22)*, Setif, Algeria.
Authors : Ahcen aliouat, Nasreddine Kouadria, Saliha Harize, Moufida Maimour
- [C4] | Multi-Threshold-based Frame Segmentation for Content-Aware Video Coding in WMSN, in *Advances in Computing Systems and Applications. CSA 2022. Lecture Notes in Networks and Systems, vol 513. Springer, Cham. Algiers, Algeria.*
Authors : Ahcen aliouat, Nasreddine Kouadria, Saliha Harize, Moufida Maimour

[Other Projects Publications : Heart Sounds classification using DL]

- [J1] | A Comparative Analysis of Constant-Q Transform, Gammatonegram, and Mel-Spectrogram Techniques for AI-aided Cardiac Diagnostics, *Medical Engineering and Physics Journal (Elsevier)*
Authors : Mohamed Sadek Mekahlia, Mohamed Fezari, Ahcen aliouat
- [C1] | Classification of Heart Sound Signals Using Scalogram and Convolutional Neural Network, *International Journal of Informatics and Applied Mathematics.*
Authors : Mohamed Sadek Mekahlia, Mohamed Fezari, Ahcen aliouat
- [C2] | CQT-based Convolutional Neural Networks for PCG Classification, 3rd *Conference on Electrical Engineering CEE'2023*
Authors : Mohamed Sadek Mekahlia, Mohamed Fezari, Ahcen aliouat

[Other Projects Publications : Video Coding]

- [J2] | A comparative evaluation of various HEVC implementations in error-prone environment, (Accepted on 20 August 2024)- *Automatic Control and Computer Sciences (Springer)*
Authors : Islem Mansri, Nasreddine Kouadria, Ahcen Aliouat, Nourededdine Doghmane

CERTIFICATES

Active reviewer for :

- IEEE ICASSP
- IEEE Transactions on Communications
- IEEE Transactions on Cognitive Communications and Networking
- Scientific Reports (Nature)
- Pattern Recognition Letters (Elsevier)
- Signal Processing Letters -IEEE (5 papers).
- IEEE Access Journal -IEEE (36 papers)
- Signal Processing - Image Communication
- Signal Processing, EURASIP-Elsevier (4 papers)
- Computer Communications
- Electronics Letters -Wiley (4 papers)
- Computers in Biology and Medicine -Elsevier
- Journal of Visual Communication and Image Representation

RESPONSIBILITIES, **Area Chair :**

SEMINARS, AND

TALKS

IEEE ICASSP 2026 in Machine Learning for Signal Processing Track

Technical Program Committee :

- ISTC (International Symposium on Topics in Coding) 2025
- The 6th Conference on Computing Systems and Applications, Algiers, Algeria
- ICECET, The 3rd and 4th International Conference on Electrical, Computer and Energy Technologies (ICECET), Sydney, Australia
- ACDSA 2024, International Conference on Artificial Intelligence, Computer, Data Sciences and Applications

Organizer :

- The 12th International Symposium on Topics in Coding, September 2023, Brest, France
- Doctoral Days and Summer School, LASA Laboratory, July 2019, Annaba, Algeria.

Seminar :

- Goal-Oriented and Semantic Communications Seminar. GdR AISIS, Paris, France.

Invited Talks :

- **Seminar & Workshop : Intelligence artificielle et compression : apprentissage à partir de données comprimées pour l'apprentissage**, à la maison de l'IA de l'Université Badji Mokhtar de Annaba - 20 Octobre 2025.
- **Machine and Deep Learning for Visual Tasks**, organized by Google Developers Group GDG Batna, and SPE Chapter group in the University of Batna 2, Algeria. 08 January 2025.
